Study Material: Fundamentals of Computer Networks

This study guide introduces the fundamental concepts of computer networking, including network types, protocols, addressing, network devices, security, and common troubleshooting techniques. It is designed as a test document for Retrieval-Augmented Generation (RAG) applications.

1. Introduction to Computer Networks

A computer network is a collection of interconnected devices that communicate and share resources. " "Devices on a network may include computers, smartphones, servers, printers, routers, and switches. " "The main purposes of networking are resource sharing, communication, centralized management, and access to services.

2. Types of Networks

A Personal Area Network (PAN) covers a very small area around an individual, such as connections between a phone and wireless headphones. " "A Local Area Network (LAN) normally covers a home, office, laboratory, or building. A Metropolitan Area Network (MAN) can connect networks across a city. " "A Wide Area Network (WAN) covers large geographical areas and can connect networks across countries or continents. The Internet is the most well-known example of a global interconnected network.

3. Network Topologies

Network topology describes how devices are arranged and connected. In a star topology, devices connect to a central switch or hub. " "In a bus topology, devices share a common communication line. A ring topology connects devices in a circular arrangement. " "A mesh topology provides multiple connections between devices and can offer high redundancy. Modern Ethernet LANs commonly use a physical star arrangement.

4. OSI Model

The Open Systems Interconnection (OSI) model divides network communication into seven layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application. " "The Physical layer deals with transmission of raw bits. The Data Link layer handles frames and local delivery. The Network layer is responsible for logical addressing and routing. " "The Transport layer provides end-to-end communication and may provide reliability. The Session layer manages communication sessions. " "The Presentation layer deals with data representation, encryption, and compression. The Application layer provides network services to applications.

5. TCP and UDP

TCP, or Transmission Control Protocol, is connection-oriented and provides reliable, ordered delivery of data. It uses acknowledgments, sequencing, and retransmission mechanisms. " "UDP, or User Datagram Protocol, is connectionless and has lower overhead. It does not guarantee delivery or ordering. " "TCP is commonly used when reliability is important, while UDP is useful for applications where speed and low latency are more important, such as some real-time communication and streaming scenarios.

6. IP Addressing

An IP address identifies a device or network interface at the network layer. IPv4 addresses contain 32 bits and are commonly written as four decimal octets, such as 192.168.1.10. " "IPv6 was developed to provide a much larger address space and uses 128-bit addresses. Private IPv4 address ranges are commonly used inside local networks, while routers can use Network Address Translation (NAT) to allow private devices to access external networks.

7. DNS

The Domain Name System (DNS) translates human-readable domain names into IP addresses. For example, a user can enter a domain name instead of remembering a numerical IP address. " "A DNS resolver queries DNS infrastructure to find the appropriate record. Common DNS records include A records for IPv4 addresses, AAAA records for IPv6 addresses, and MX records for mail servers.

8. DHCP

Dynamic Host Configuration Protocol (DHCP) automatically provides network configuration information to clients. " "A DHCP server can assign an IP address, subnet mask, default gateway, and DNS server information. " "The basic DHCP process is often described using DORA: Discover, Offer, Request, and Acknowledge.

9. Network Devices

A switch connects devices within a local network and forwards Ethernet frames based on MAC addresses. " "A router connects different networks and forwards packets based on IP addresses. A modem provides an interface between a local network and certain types of service-provider infrastructure. " "An access point provides wireless connectivity to network clients. A firewall monitors and controls traffic according to security rules.

10. HTTP and HTTPS

HTTP is an application-layer protocol used to transfer web resources between clients and servers. HTTPS is HTTP protected by Transport Layer Security (TLS). " "TLS helps provide confidentiality, integrity, and authentication. When a browser connects to an HTTPS website, the server presents a digital certificate that helps establish trust and a secure encrypted connection.

11. Network Security

Network security protects systems and data from unauthorized access, disruption, and misuse. Common security measures include firewalls, strong authentication, encryption, access control, software updates, and network monitoring. " "Phishing is a social-engineering technique in which an attacker attempts to trick a person into revealing information or performing an unsafe action. " "Using multi-factor authentication can reduce the impact of stolen passwords.

12. Common Troubleshooting

A systematic troubleshooting process can save time. First identify the symptoms and determine whether the problem affects one device or many devices. " "Check physical connections and wireless status. Verify IP configuration and test local connectivity. " "Useful diagnostic tools include ping for connectivity testing, tracert or traceroute for examining paths, and nslookup or dig for DNS queries. " "Testing one layer at a time helps isolate the cause of a problem.

Review Questions

1. What is the main purpose of a computer network?
2. How does a LAN differ from a WAN?
3. Which topology uses a central device?
4. Name the seven layers of the OSI model.
5. What is the main difference between TCP and UDP?
6. How many bits are contained in an IPv4 address?
7. What problem does DNS solve?
8. What does DHCP provide to network clients?
9. What is the difference between a switch and a router?
10. Why is HTTPS more secure than ordinary HTTP?
11. What are three common network-security measures?
12. What is phishing?
13. Which tool can be used to test basic network connectivity?
14. What does NAT commonly allow private IPv4 devices to do?
15. Why is systematic troubleshooting useful?

Quick Reference Table

Concept	Key Point
LAN	Network covering a limited area such as a home or office.
WAN	Network spanning large geographical areas.
TCP	Reliable, connection-oriented transport protocol.
UDP	Connectionless transport protocol with low overhead.
DNS	Maps domain names to network addresses.
DHCP	Automatically assigns network configuration.
Switch	Forwards local Ethernet frames using MAC addresses.
Router	Connects networks and forwards IP packets.
HTTPS	HTTP protected using TLS.
Firewall	Controls traffic according to security rules.